Contents

Project 3 report

Zhen Lai 10/3/2024


```
repo_dir = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
addpath(fullfile(repo_dir, 'shared'));
paths = imgs351Paths(repo_dir);
addpath(paths.p03.code);
```

step 2 - define function loadCIEdata

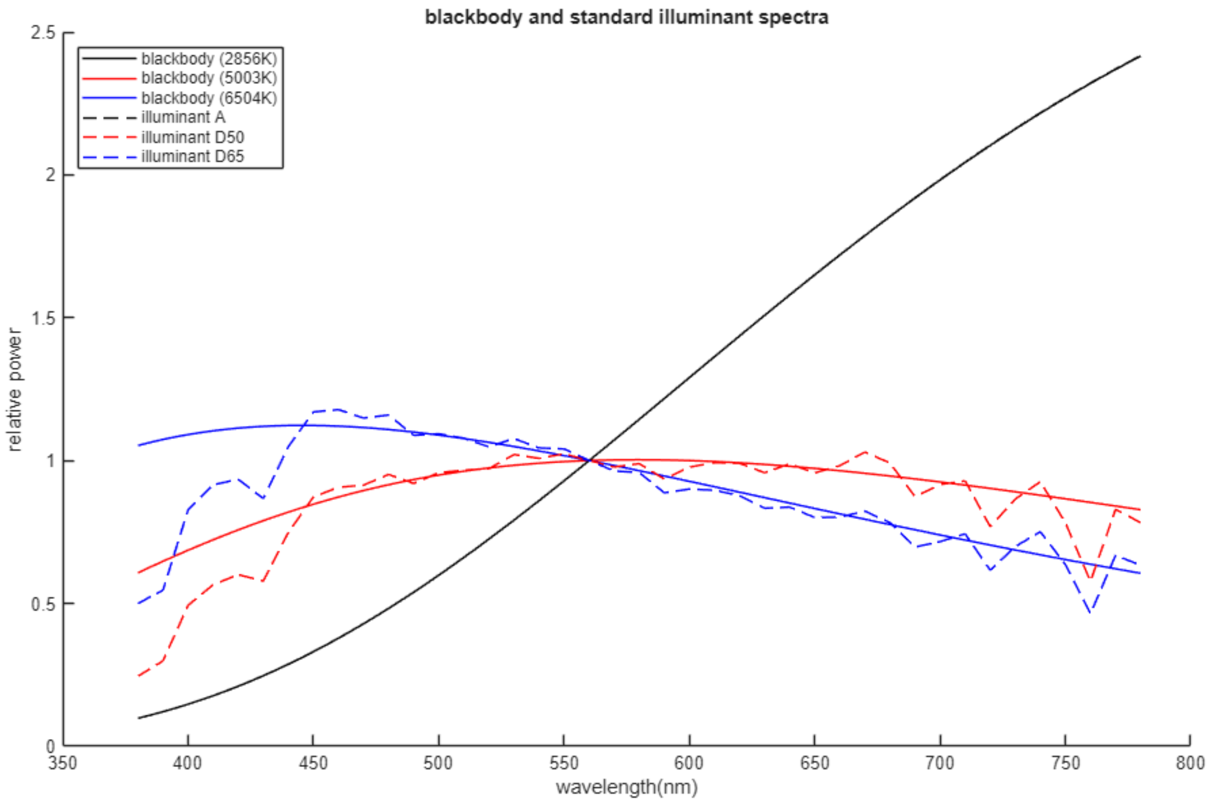
Define a MATLAB function that returns a structure of CIE observer and illuminant data. Below is the function.

```
function cie = loadCIEdata
%LOADCIEDATA Load the CIE observer and illuminant data used in this course.
%   CIE = LOADCIEDATA returns a structure containing wavelength samples,
%   2-degree and 10-degree color-matching functions, standard illuminants,
%   and a perfect reflecting diffuser.

project_dir = fileparts(fileparts(mfilename('fullpath')));
resource_dir = fullfile(project_dir, 'data');

cie = struct();

% import 1931 standard observer data
cie_2deg = importdata(fullfile(resource_dir, 'CIE_2Deg_380-780-5nm.txt'));
% store the wavelength data from cie_2deg to the lambda field
cie.lambda = cie_2deg(:, 1);
cie.cmf2deg = cie_2deg(:, 2:end);

% Import the 1964 10-degree standard observer.
cie_10deg = importdata(fullfile(resource_dir, 'CIE_10Deg_380-780-5nm.txt'));
cie.cmf10deg = cie_10deg(:, 2:end);

% import standard illuminant A data
illA = importdata(fullfile(resource_dir, 'CIE_IllA_380-780-5nm.txt'));
cie.illA = illA(:, 2:end);

% import standard illuminant C data
illC = importdata(fullfile(resource_dir, 'CIE_IllC_380-780-5nm.txt'));
cie.illC = illC(:, 2:end);

% import standard illuminant D50 data
illD50 = importdata(fullfile(resource_dir, 'CIE_IllD50_380-780-5nm.txt'));
cie.illD50 = illD50(:, 2:end);

% import standard illuminant D65 data
illD65 = importdata(fullfile(resource_dir, 'CIE_IllD65_380-780-5nm.txt'));
cie.illD65 = illD65(:, 2:end);

sampleCount = numel(cie.lambda);

% Equal-energy illuminant E.
cie.illE = 100 * ones(sampleCount, 1);

% import standard illuminant F 1-12 data
illF = importdata(fullfile(resource_dir, 'CIE_IllF_1-12_380-780-5nm.txt'));
cie.illF = illF(:, 2:end);

% Perfect reflecting diffuser.
cie.PRD = ones(sampleCount, 1, 'double');
```

```
end
```

step 3a - plot blackbody and standard illuminant spectra

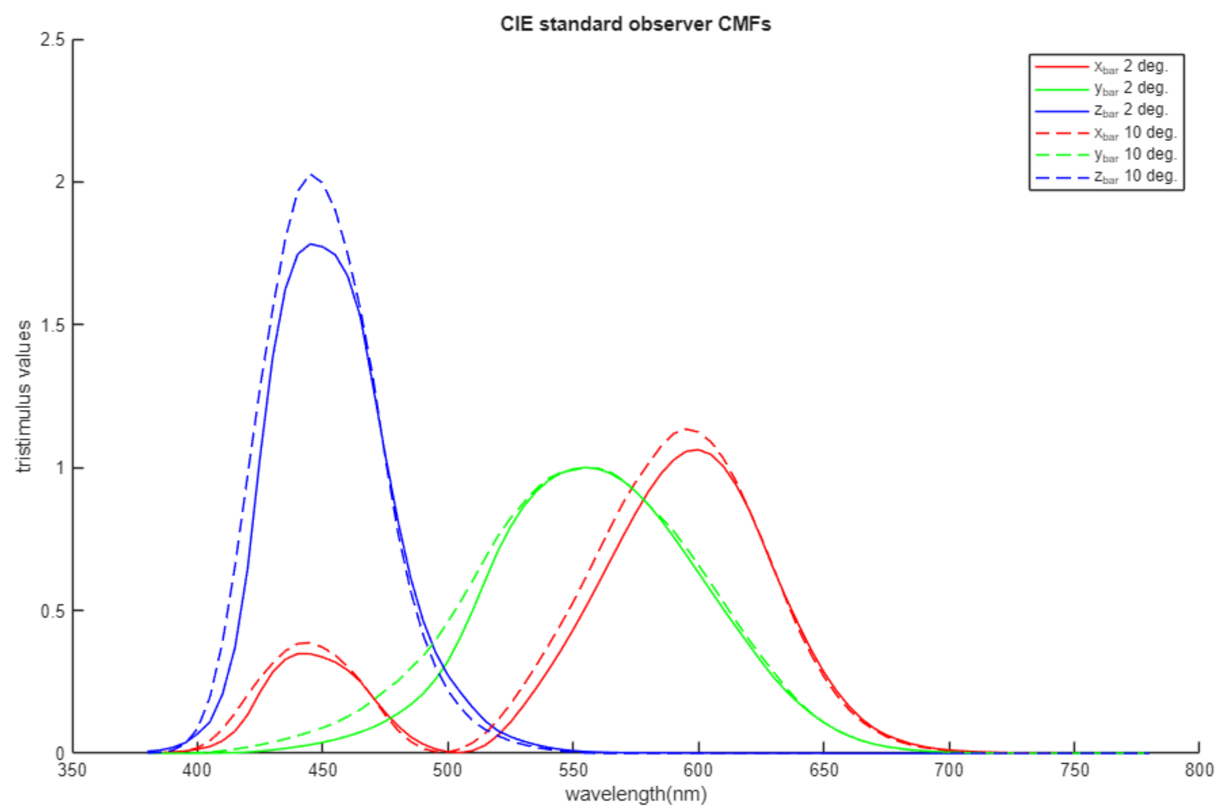
```
% blackbody and standard illuminant spectra
% normalizing standard illuminant SPD data to 1.0 at 560nm
cie = loadCIEdata;
illA = transpose(0.01 * cie.illA);
illD50 = transpose(0.01 * cie.illD50);
illD65 = transpose(0.01 * cie.illD65);

% set x axis
xs = transpose(cie.lambda);

figure;
hold on;
% blackbody (2856K)
plot(xs,blackbody(2856, xs), 'k');

% blackbody (5003K)
plot(xs,blackbody(5003, xs), 'r');

% blackbody 6504K
plot(xs,blackbody(6504, xs), 'b');

% illuminant A
plot(xs, illA, '--k');

% illuminant D50
plot(xs, illD50, '--r');

% illuminant D65
plot(xs, illD65, '--b');
hold off;

xlim([350 800])
ylim([0 2.5])
yticks(0:(2.5/5):2.5)
xlabel('wavelength(nm)')
ylabel('relative power')
legend({'blackbody (2856K)','blackbody (5003K)','blackbody (6504K)', ...
    'illuminant A', 'illuminant D50', 'illuminant D65'}, ...
    'Location','northwest');
title('blackbody and standard illuminant spectra')
```



step 3b - plot CIE standard observer CMFs

```
% CIE standard observer CMFs

% set x axis
xs = transpose(cie.lambda);

% get cmf 2 deg. data
cmf2deg_x = transpose(cie.cmf2deg(:,1));
cmf2deg_y = transpose(cie.cmf2deg(:,2));
cmf2deg_z = transpose(cie.cmf2deg(:,3));

% get cmf 10 deg. data
cmf10deg_x = transpose(cie.cmf10deg(:,1));
cmf10deg_y = transpose(cie.cmf10deg(:,2));
cmf10deg_z = transpose(cie.cmf10deg(:,3));

figure;
hold on;

% x_bar 2 deg.
plot(xs, cmf2deg_x, 'r')

% y_bar 2 deg.
plot(xs, cmf2deg_y, 'g')

% z_bar 2 deg.
plot(xs, cmf2deg_z, 'b')

% x_bar 10 deg.
plot(xs, cmf10deg_x, '--r')

% y_bar 10 deg.
plot(xs, cmf10deg_y, '--g')

% z_bar 10 deg.
plot(xs, cmf10deg_z, '--b')
hold off;

xlim([350 800])
ylim([0 2.5])
yticks(0:(2.5/5):2.5)
xlabel('wavelength(nm)')
ylabel('tristimulus values')
legend('x_{bar} 2 deg.','y_{bar} 2 deg.','z_{bar} 2 deg.', ...
```

```
    'x_{bar} 10 deg.','y_{bar} 10 deg.','z_{bar} 10 deg.');
title('CIE standard observer CMFs')
```



step 4 - define ref2XYZ function

Define a MATLAB function that returns XYZ tristimulus values from surface reflectance, color matching function and illumination data. Below is the function.

```
function XYZ = ref2XYZ(reflectances, colorMatchingFunctions, illuminant)
%REF2XYZ Convert reflectance spectra to CIE XYZ tristimulus values.
%   XYZ = REF2XYZ(REFLECTANCES, COLORMATCHINGFUNCTIONS, ILLUMINANT)
%   accepts one or more reflectance spectra in an N-by-M matrix, the CIE
%   color-matching functions in an N-by-3 matrix, and an N-by-1 illuminant
%   spectral power distribution. The result is a 3-by-M matrix scaled so a
%   perfect reflecting diffuser has Y = 100.

normalization = 100 / sum(colorMatchingFunctions(:, 2) .* illuminant);
weightedReflectances = illuminant .* reflectances;
XYZ = normalization * colorMatchingFunctions.' * weightedReflectances;
end
```

step 5 - calculate ColorChecker XYZ values

```
% load CIE observer and illuminant data
cie = loadCIEdata;

% load color check reflectance data
cc_spectra = importdata(fullfile(paths.p03.data, ...
    'ColorChecker_380_780_5nm.txt'));

% calculate XYZ tristimulus values for each patch in the color checker
CC_XYZs = zeros(3, 24);
for patch_num = 2:25
    CC_XYZs(:, patch_num-1) = ref2XYZ(cc_spectra(:, patch_num), cie.cmf2deg, cie.illD65);
end

% display the XYZ tristimulus values
CC_XYZs
```

```
CC_XYZs =

  Columns 1 through 7
```

```
   11.5145   39.1346   18.3488   11.1492   25.8437   31.7110   37.1457
   10.3819   36.5981   19.6332   13.8551   24.3868   43.8600   29.5592
    7.1502   27.0564   35.6470    7.4267   45.6142   44.8778    6.5006

  Columns 8 through 14

   13.8627   29.1328    8.5889   33.9174   46.1864    8.9183   15.0353
   12.3179   19.8475    6.4569   44.1533   42.4957    6.4177   24.1079
   39.3093   14.9941   15.4745   11.4297    8.6771   32.2736    9.6379

  Columns 15 through 21

   19.3447   55.8457   29.6768   14.4138   87.8402   57.9621   35.2286
   11.3576   58.9726   19.3515   19.9750   92.3781   61.0426   37.0414
    5.5526    9.6411   32.2626   39.0008   95.6125   65.4909   40.2256

  Columns 22 through 24

   19.3492    8.7646    3.2111
   20.4708    9.2915    3.3763
   22.1545   10.3188    3.9312
```

step 6 - define XYZ2xyY function

Define a MATLAB function that returns the x, y chromaticity coordinates and Y luminance factor from XYZ tristimulus values

```
function xyY = XYZ2xyY(XYZ)
%XYZ2XYY Convert CIE XYZ values to x, y chromaticity and luminance.
%   XYY = XYZ2XYY(XYZ) accepts a 3-by-N matrix and returns a 3-by-N
%   matrix whose rows contain x, y, and Y. Chromaticity is undefined when
%   X + Y + Z is zero, so x and y are returned as NaN for those samples.

tristimulusSum = sum(XYZ, 1);
xyY = [XYZ(1, :) ./ tristimulusSum;
       XYZ(2, :) ./ tristimulusSum;
       XYZ(2, :)];
end
```

step 7 - calculate ColorChecker xyY values

```
CC_xyYs = XYZ2xyY(CC_XYZs);

% display the xyY values
CC_xyYs
```

```
CC_xyYs =

  Columns 1 through 7

    0.3964    0.3807    0.2492    0.3438    0.2696    0.2633    0.5074
    0.3574    0.3561    0.2667    0.4272    0.2544    0.3641    0.4038
   10.3819   36.5981   19.6332   13.8551   24.3868   43.8600   29.5592

  Columns 8 through 14

    0.2117    0.4554    0.2814    0.3790    0.4744    0.1873    0.3082
    0.1881    0.3102    0.2116    0.4933    0.4365    0.1348    0.4942
   12.3179   19.8475    6.4569   44.1533   42.4957    6.4177   24.1079

  Columns 15 through 21

    0.5336    0.4487    0.3651    0.1964    0.3185    0.3142    0.3132
    0.3133    0.4738    0.2381    0.2722    0.3349    0.3309    0.3293
   11.3576   58.9726   19.3515   19.9750   92.3781   61.0426   37.0414

  Columns 22 through 24

    0.3122    0.3089    0.3053
    0.3303    0.3275    0.3210
   20.4708    9.2915    3.3763
```

step 8 - load painted patches spectra data

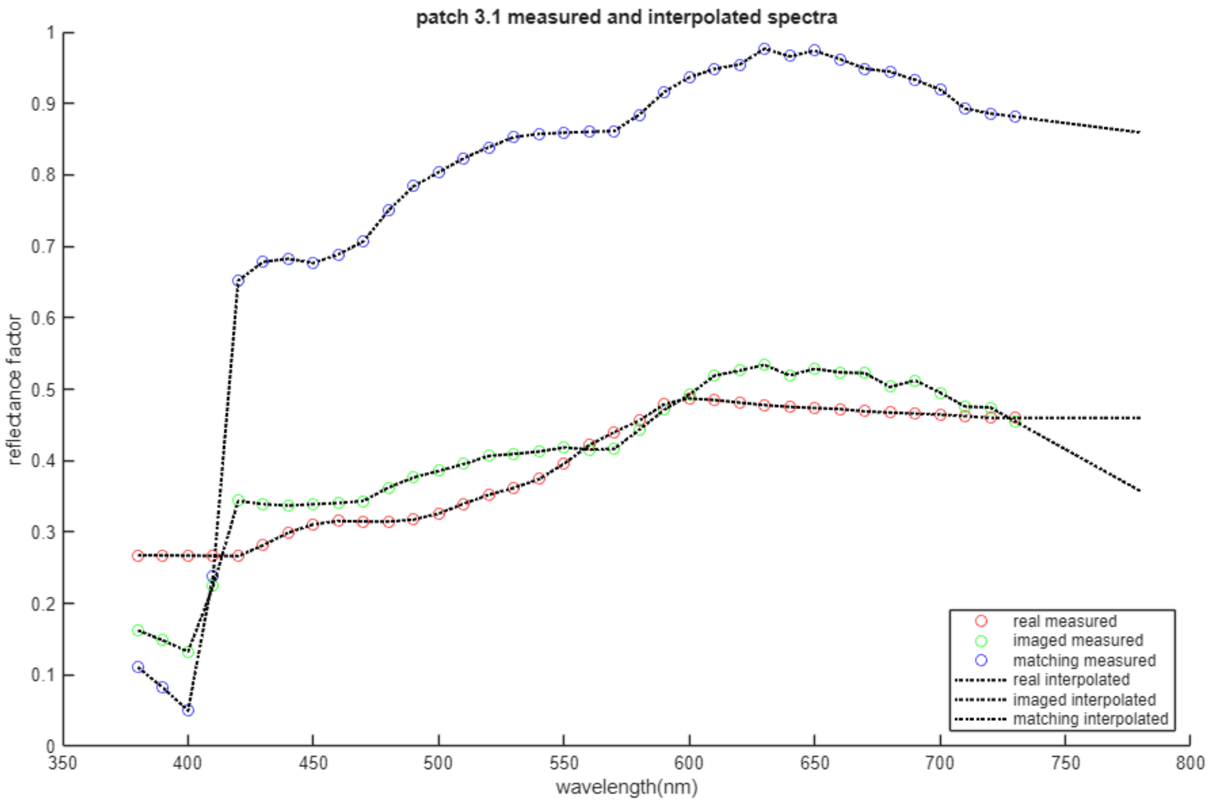
```
% load the CIE observer and illuminant data
cie = loadCIEdata;

% define ColorMunki/Argyll/spotread measurement wavelengths
cm_lams = 380:10:730;
```

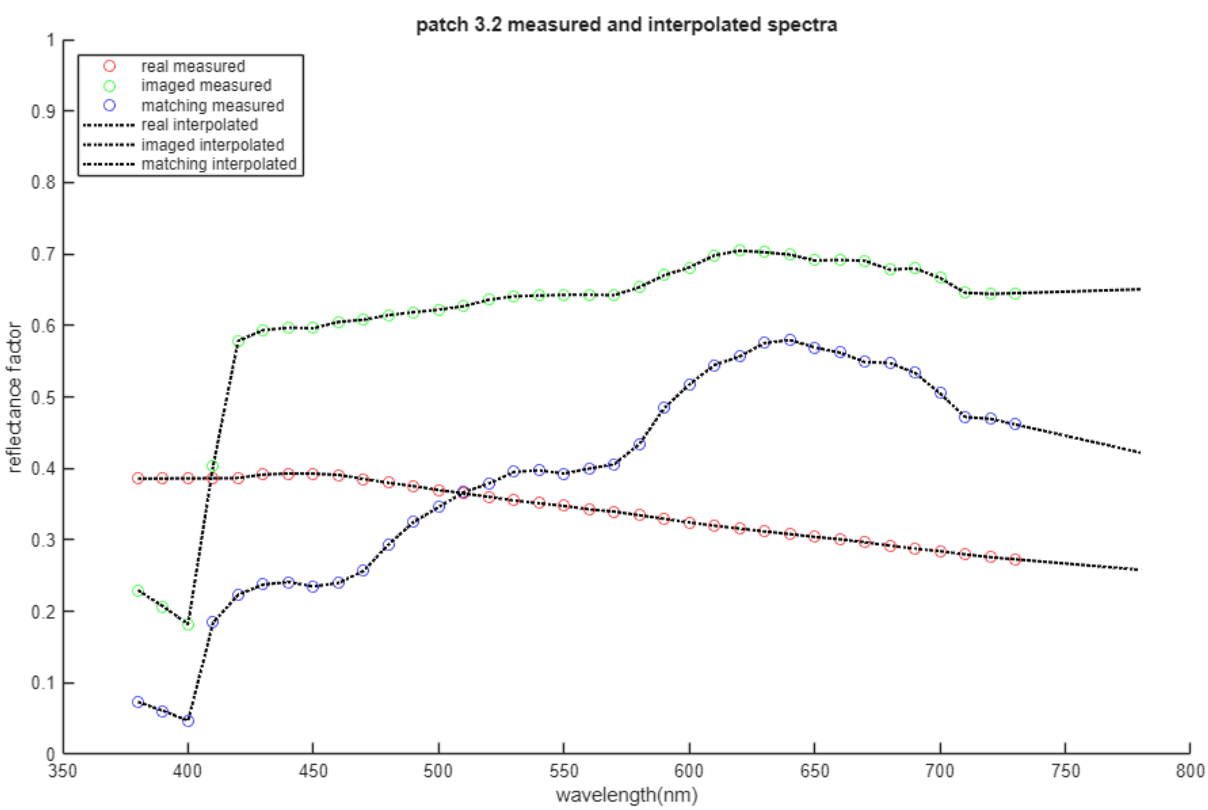
```
% define header offsets for reading the .sp files
cm_g_offset_spotread = 19;
cm_g_offset_spotreadt = 18;

% load and normalize the measured spectral data for the patch 3.1
data_real31 = importdata(fullfile(paths.p02.data, 'colormunki_data', ...
    'real', '3.1_real.sp'), ...
    ' ', cm_g_offset_spotread);
real_31 = data_real31.data/100;

data_imaged31 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'imaged', ...
    '3.1_imaged.sp'), ' ', ...
    cm_g_offset_spotreadt);
imaged_31 = data_imaged31.data/100;

data_matching31 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'matched', ...
    '3.1_matching.sp'), ' ', ...
    cm_g_offset_spotreadt);
matching_31 = data_matching31.data/100;

% load and normalize the measured spectral data for the patch 3.2
data_real32 = importdata(fullfile(paths.p02.data, 'colormunki_data', ...
    'real', '3.2_real.sp'), ...
    ' ', cm_g_offset_spotread);
real_32 = data_real32.data/100;

data_imaged32 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'imaged', ...
    '3.2_imaged.sp'), ' ', ...
    cm_g_offset_spotreadt);
imaged_32 = data_imaged32.data/100;

data_matching32 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'matched', ...
    '3.2_matching.sp'), ' ', ...
    cm_g_offset_spotreadt);
matching_32 = data_matching32.data/100;
```

step 9 - plot the measured and interpolated spectra for paint patches

```
% INTERPOLATE/EXTRAPOLATE DATA

% patch 3.1
interpolated_real_31 = interp1(cm_lams, transpose(real_31), ...
    cie.lambda(:), 'linear', 'extrap');
interpolated_imaged_31 = interp1(cm_lams, transpose(imaged_31), ...
    cie.lambda(:), 'linear', 'extrap');
interpolated_matching_31 = interp1(cm_lams, transpose(matching_31), ...
    cie.lambda(:), 'linear', 'extrap');

% patch 3.2
interpolated_real_32 = interp1(cm_lams, transpose(real_32), ...
    cie.lambda(:), 'linear', 'extrap');
interpolated_imaged_32 = interp1(cm_lams, transpose(imaged_32), ...
    cie.lambda(:), 'linear', 'extrap');
interpolated_matching_32 = interp1(cm_lams, transpose(matching_32), ...
    cie.lambda(:), 'linear', 'extrap');


% PLOTTING INTERPOLATED SPECTRA
% set x axes
measured_xs = cm_lams;
interpolated_xs =  transpose(cie.lambda);

% measured and interpolated spectra patch 3.1
figure;
hold on;
% measured
plot(measured_xs, real_31, 'or', 'LineWidth', 0.5)
plot(measured_xs, imaged_31, 'og', 'LineWidth', 0.5)
plot(measured_xs, matching_31, 'ob', 'LineWidth', 0.5)

% interpolated
plot(interpolated_xs, interpolated_real_31, ':', 'Color', 'black', 'LineWidth', 1.5)
plot(interpolated_xs, interpolated_imaged_31, ':', 'Color', 'black', 'LineWidth', 1.5)
plot(interpolated_xs, interpolated_matching_31, ':', 'Color', 'black', 'LineWidth', 1.5)
hold off

xlim([350 800])
ylim([0 1])
yticks(0:(1/10):1)
xlabel('wavelength(nm)')
```

```matlab
ylabel('reflectance factor')
legend({'real measured', 'imaged measured', 'matching measured', ['real ' ...
    'interpolated'], 'imaged interpolated', 'matching interpolated'}, ...
    'Location', 'southeast');
title('patch 3.1 measured and interpolated spectra')

% measured and interpolated spectra patch 3.2
figure;
hold on;
% measured
plot(measured_xs, real_32, 'or', 'LineWidth', 0.5)
plot(measured_xs, imaged_32, 'og', 'LineWidth', 0.5)
plot(measured_xs, matching_32, 'ob', 'LineWidth', 0.5)

% interpolated
plot(interpolated_xs, interpolated_real_32, ':', 'Color', 'black', 'LineWidth', 1.5)
plot(interpolated_xs, interpolated_imaged_32, ':', 'Color', 'black', 'LineWidth', 1.5)
plot(interpolated_xs, interpolated_matching_32, ':', 'Color', 'black', 'LineWidth', 1.5)
hold off

xlim([350 800])
ylim([0 1])
yticks(0:(1/10):1)
xlabel('wavelength(nm)')
ylabel('reflectance factor')
legend({'real measured', 'imaged measured', 'matching measured', ['real ' ...
    'interpolated'], 'imaged interpolated', 'matching interpolated'}, ...
    'Location', 'northwest');
title('patch 3.2 measured and interpolated spectra')
```





step 10 - list measured and calculated XYZ values for paint patches

```
% CALCULATING XYZs
% calculate the tristimulus values and chromaticity coordinates
% for patch 3.1
XYZ_real_31 = ref2XYZ(interpolated_real_31, cie.cmf2deg, cie.illD50);
```

```
XYZ_real_31 = reshape(XYZ_real_31, 3, 1);

XYZ_imaged_31 = ref2XYZ(interpolated_imaged_31, cie.cmf2deg, cie.illD50);
XYZ_imaged_31 = reshape(XYZ_imaged_31, 3, 1);

XYZ_matching_31 = ref2XYZ(interpolated_matching_31, cie.cmf2deg, cie.illD50);
XYZ_matching_31 = reshape(XYZ_matching_31, 3, 1);

% calculate the tristimulus values and chromaticity coordinates
% for patch 3.2
XYZ_real_32 = ref2XYZ(interpolated_real_32, cie.cmf2deg, cie.illD50);
XYZ_real_32 = reshape(XYZ_real_32, 3, 1);

XYZ_imaged_32 = ref2XYZ(interpolated_imaged_32, cie.cmf2deg, cie.illD50);
XYZ_imaged_32 = reshape(XYZ_imaged_32, 3, 1);

XYZ_matching_32 = ref2XYZ(interpolated_matching_32, cie.cmf2deg, cie.illD50);
XYZ_matching_32 = reshape(XYZ_matching_32, 3, 1);

% LISTING MEASURED AND CALCULATED XYZs
% load measured data
load(fullfile(paths.p02.data, 'patchColorimetryData.mat'))

% print title
fprintf('\n%s\n', "Measured and calculated tristimulus values")

% print first table for patch 3.1
fprintf('\n                              %8s\n', 'patch 3.1')
fprintf('                   %8s                    %10s \n', 'measured', 'calculated')
fprintf('%5s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', 'X', 'Y', 'Z', 'X', 'Y', 'Z')
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'real', ...
    XYZLabsreal(1,2:4), XYZ_real_31)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'imaged', ...
    XYZLabsimaged(1,2:4), XYZ_imaged_31)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'matching', ...
    XYZLabsmatching(1,2:4), XYZ_matching_31)

% print second table for patch 3.2
fprintf('\n                              %8s\n', 'patch 3.2')
fprintf('                   %8s                    %10s \n', 'measured', 'calculated')
fprintf('%5s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', 'X', 'Y', 'Z', 'X', 'Y', 'Z')
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'real', ...
    XYZLabsreal(2,2:4), XYZ_real_32)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'imaged', ...
    XYZLabsimaged(2,2:4), XYZ_imaged_32)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'matching', ...
    XYZLabsmatching(2,2:4), XYZ_matching_32)
```

```
Measured and calculated tristimulus values

                              patch 3.1
                   measured                    calculated
               X         Y         Z         X         Y         Z
    real   42.2036   41.3758   25.3596   42.1870   41.3717   25.3597
  imaged   44.1598   43.6064   28.4608   44.1485   43.6078   28.4532
matching   85.2686   87.3024   57.3942   85.2451   87.2978   57.3463

                              patch 3.2
                   measured                    calculated
               X         Y         Z         X         Y         Z
    real   32.3292   34.2553   31.9201   32.3267   34.2556   31.9201
  imaged   63.9017   65.2199   49.3994   63.8858   65.2197   49.3682
matching   44.0165   42.7388   21.1155   44.0068   42.7391   21.1241
```

step 11 - list measured and calculated chromaticity coordinates for paint patches

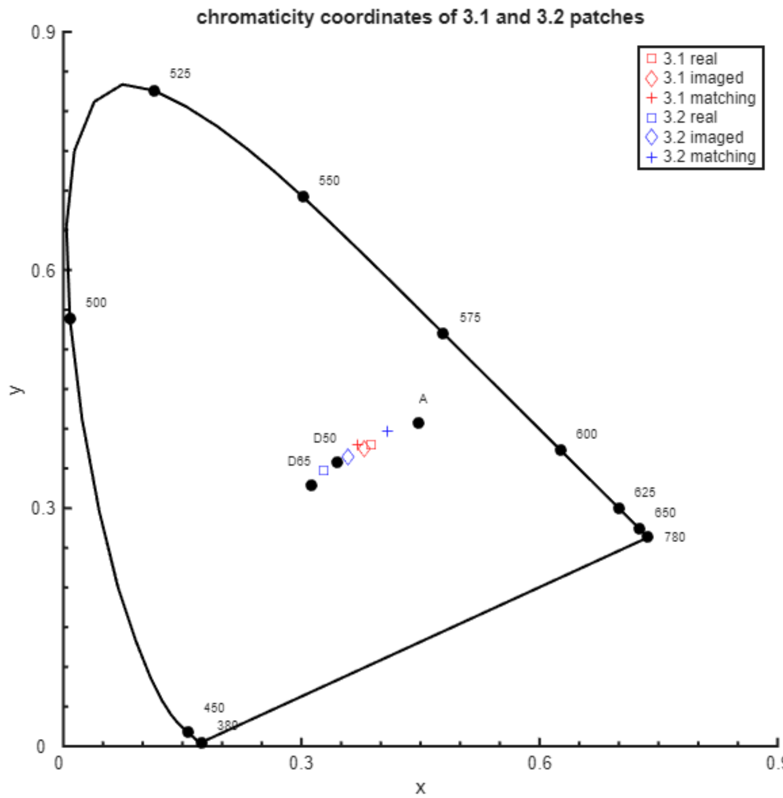
```
% CALCULATING xyY
% xyY from calculated XYZ
xyY_real_31 = XYZ2xyY(XYZ_real_31);
xyY_imaged_31 = XYZ2xyY(XYZ_imaged_31);
xyY_matching_31 = XYZ2xyY(XYZ_matching_31);

xyY_real_32 = XYZ2xyY(XYZ_real_32);
xyY_imaged_32 = XYZ2xyY(XYZ_imaged_32);
xyY_matching_32 = XYZ2xyY(XYZ_matching_32);

% xyY from measured XYZ
xyY_real_31_measured = XYZ2xyY(XYZLabsreal(1,2:4)');
xyY_imaged_31_measured = XYZ2xyY(XYZLabsimaged(1,2:4)');
xyY_matching_31_measured = XYZ2xyY(XYZLabsmatching(1,2:4)');

xyY_real_32_measured = XYZ2xyY(XYZLabsreal(2,2:4)');
xyY_imaged_32_measured = XYZ2xyY(XYZLabsimaged(2,2:4)');
```

```
xyY_matching_32_measured = XYZ2xyY(XYZLabsmatching(2,2:4)');

% LISTING xyY
% print title
fprintf('\n%s\n', "Measured and calculated chromaticity coordinates")

% print first table for patch 3.1
fprintf('\n%8s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', '', '','patch 3.1', '', '', '')
fprintf('%8s  %8s  %8s  %8s  %8s  %10s  %8s\n', '', '', 'measured','', '', 'calculated', '')
fprintf('%5s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', 'x', 'y', 'Y', 'x', 'y', 'Y')
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'real', ...
    xyY_real_31_measured, xyY_real_31)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'imaged', ...
    xyY_imaged_31_measured, xyY_imaged_31)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'matching', ...
    xyY_matching_31_measured, xyY_matching_31)

% print second table for patch 3.2
fprintf('\n%8s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', '', '','patch 3.2', '', '', '')
fprintf('%8s  %8s  %8s  %8s  %8s  %10s  %8s\n', '', '', 'measured','', '', 'calculated', '')
fprintf('%5s  %8s  %8s  %8s  %8s  %8s  %8s\n', '', 'x', 'y', 'Y', 'x', 'y', 'Y')
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'real', ...
    xyY_real_32_measured, xyY_real_32)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'imaged', ...
    xyY_imaged_32_measured, xyY_imaged_32)
fprintf('%8s  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f  %8.4f\n', 'matching', ...
    xyY_matching_32_measured, xyY_matching_32)
```

```
Measured and calculated chromaticity coordinates

                              patch 3.1
                    measured                      calculated
               x         y         Y          x         y         Y
    real    0.3874    0.3798   41.3758    0.3873    0.3798   41.3717
  imaged    0.3799    0.3752   43.6064    0.3799    0.3753   43.6078
matching    0.3708    0.3796   87.3024    0.3708    0.3797   87.2978

                              patch 3.2
                    measured                      calculated
               x         y         Y          x         y         Y
    real    0.3282    0.3478   34.2553    0.3282    0.3478   34.2556
  imaged    0.3580    0.3653   65.2199    0.3580    0.3654   65.2197
matching    0.4080    0.3962   42.7388    0.4080    0.3962   42.7391
```

step 12 - visualize chromaticity coordinates of paint patches on chromaticity diagram

```
% plot chromaticity diagram
plot_chrom_diag_skel;
hold on;

% plot patch 3.1 calculated chromaticity coordinates
plot(xyY_real_31(1,1),xyY_real_31(2,1),'square','Color', 'red')
plot(xyY_imaged_31(1,1),xyY_imaged_31(2,1),'diamond','Color', 'red')
plot(xyY_matching_31(1,1),xyY_matching_31(2,1),'+','Color', 'red')

% plot patch 3.2 calculated chromaticity coordinates
plot(xyY_real_32(1,1),xyY_real_32(2,1),'square','Color', 'blue')
plot(xyY_imaged_32(1,1),xyY_imaged_32(2,1),'diamond','Color', 'blue')
plot(xyY_matching_32(1,1),xyY_matching_32(2,1),'+','Color', 'blue')

fontsize(gcf, scale=0.7)

hold off

%legend({'3.1 real', '3.1 imaged', '3.1 matching', '3.2 real', ...
    %'3.2 imaged', '3.2 matching'}, 'Location', 'northeast');
legend({'','','','','','','','','','','','','','', '3.1 real', ...
    '3.1 imaged', '3.1 matching', '3.2 real', '3.2 imaged',  ...
    '3.2 matching'}, 'Location', 'northeast');
title('chromaticity coordinates of 3.1 and 3.2 patches')
```



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